

PRADEEP SINGH

SENIOR EMBEDDED ENGINEER | AUTOSAR SPECIALIST

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SUMMARY

Senior Embedded software engineer with 12+ years of experience architecting, configuring, and integrating safety-critical embedded platforms for global OEMs (BMW, JLR, VW, Volvo). Expert in classic AUTOSAR (SWC, BSW, MCAL, RTE), ISO 26262 alignment, and multi-ECU platform strategy. Proven leader in defining AUTOSAR architecture standards, optimizing RTE/BSW/MCAL performance, and guiding cross-global teams from concept to SOP.

PROFESSIONAL EXPERIENCE

Luxoft GmbH | Munich, Germany

DEC 2019 - PRESENT

Senior AUTOSAR Platform Architect / Domain Lead

Project: BMW Body & Gateway ECUs

Language: Embedded C/C++

Responsibilities:

- Defined and standardized classic AUTOSAR Platform architecture (MCAL → BSW → RTE → SWC) reused across multiple BMW ECU programs.
- Led configuration and integration of COM(CAN/LIN), DIAG (DCM/DEM), RTE & MCAL modules.
- Implemented board bring-up and startup code initialization for ARM-based gateway ECUs, configuring clock trees, memory sections, and interrupt vectors.
- Implemented memory footprint optimization strategies reducing stack usage by ~12% across platform modules.
- Aligned AUTOSAR architecture with ISO 26262 ASIL-B safety requirements, ensuring full bidirectional traceability of 300+ safety and system requirements.
- Participated in FMEA/FMEDA discussions for safety mechanism validation.
- Optimized RTE task mapping and scheduling, reducing CPU load by 18% under peak communication conditions.
- Acted as AUTOSAR Technical Authority reviewing 200+ change requests for architectural compliance.
- Development in C/C++ with strict MISRA compliance, improving software reliability and maintainability.
- Led integration and validation campaigns executing 200+ integration test cases, achieving 98% requirement coverage.
- Defined and enforced unit testing strategy (GTest/GMock) integrated into CI/CD, reducing regression defects by 20%.
- Performed multi-core debugging using Lauterbach TRACE32, accelerating root-cause analysis and reducing issue resolution time by 30%.
- Mentored 12+ engineers in AUTOSAR configuration strategies, BSW debugging, and integration best practices.

Lear Automotive | Pune, India

JUL 2017 - NOV 2019

Technical Specialist

Project: JLR BISG Inverter ECU & X590 Hybrid EV Charger

Language: Embedded C

Responsibilities:

- Led AUTOSAR BSW and Application architecture alignment to ISO 26262 ASIL-B requirements, ensuring 100% bidirectional traceability between 300+ safety requirements and software components.
- Configured and Integrated COM (CAN/FlexRay), DIAG (DCM/DEM), and NVM modules aligned with safety goals, enabling detection and reporting of 50+ safety-relevant events.
- Integrated Bootloader and application software ensuring compliant reprogramming sequence per OEM standards.
- Developed custom secondary bootloader supporting UART-based firmware updates, accelerating validation cycles by 60%.
- Designed linker scripts and optimized memory mapping to support dual-image (bootloader + application) architecture.
- Configured and integrated AUTOSAR MCAL drivers (SPI, I2C, ADC, PWM, DIO) for real-time signal acquisition and peripheral communication.
- Conducted integration validation on HIL platforms achieving >95% test coverage.
- Enforced coding guidelines and AUTOSAR best practices across distributed teams of 8+ engineers.

Lear Automotive | Pune, India

AUG 2016 - JUN 2017

Senior Software Engineer

Project: VOLVO HCM Gen2 Headlight Control Module

Language: Embedded C

Responsibilities:

- Configured & Integrated AUTOSAR CAN stack in DaVinci Configurator.
- Configured AUTOSAR RTE for signal mapping between Application and BSW.
- Ensured software quality with Unit Tests, Integration Tests and static code analysis.
- Designed and executed automated integration tests, achieving high defect containment before system-level testing.
- Performed board-level debugging and in-vehicle validation, reducing field issues during production release.

Lear Automotive | Pune, India

APR 2013 - JUL 2016

Software Engineer

Project: Volkswagen BCM

Language: Embedded C/C++

Responsibilities:

- Analyzed customer system requirements and derived structured software specifications.
- Developed Application software for Exterior/Interior Lights & Wiper/Washer System following the AUTOSAR 4.2 standards.
- Configured RTE mappings and validated signal interfaces between Application and BSW.
- Supported ISO 26262 safety concept alignment and safety reviews.
- Modeled control logic using MATLAB/Simulink and generated production-ready C/C++ code..
- Ensured software quality with Unit Tests, Integration Tests and MISRA C compliance.
- Debugged and validated software on Test bench/HIL and In-Vehicle.
- Performed design/code review tasks.
- Provided technical direction to team members in achieving 100% coverage of requirements.
- Supported Engineering team for software specific technical solutions and present to the OEM.

Project: Freescale Automotive MCUs

Language: Embedded C

Responsibilities:

- Developed low-level device driver initialization code for MPC56xx series microcontrollers.
- Validated tool-generated peripheral configuration code on target hardware.
- Performed multi-core debugging using Lauterbach debugger.

KEY ACHIEVEMENTS & TECHNICAL INNOVATIONS

- Designed and implemented a reusable ECU Test Automation Framework adopted across 3 projects, reducing validation cycle time by 50% (**€60,000 annual savings**).
- Developed CAN/LIN/JTAG Switcher (Python + custom PCB) resulting in hardware **cost savings upto €10000** in project cost and improved remote debugging flexibility across global teams.
- Developed a Master Bootloader Emulator that **accelerated validation by 60%** and eliminated the need for manual testing.

CORE COMPETENCIES

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|---|-----------------------------------|------------------------------|
| • Classic AUTOSAR Architecture (4.x) | • Embedded C / C++ | • Board Bringup |
| • SWC / RTE / BSW / MCAL Configuration | • CAN, LIN, FlexRay, Ethernet | • CI/CD & DevOps Integration |
| • ISO 26262 Functional Safety Alignment | • Bootloader | • Jira & Confluence |
| • COM / DCM / DEM / NVM / WDG | • ARM Cortex / PowerPC | • Scrum / Agile |
| | • Vector DaVinci, EB Tresos, KSAR | |
| | • CI/CD & Quality Gates | |
| | • ASPICE | |

TOOLS & TECHNOLOGIES

- **Programming:** Embedded C, C++11/14
- **RTOS:** FreeRTOS, AUTOSAR OS, Real-Time Scheduling
- **Microcontrollers / SoC:** ARM Cortex-M, NXP S32G3, Stellar G7, MPC57xx, RH850
- **Bootloaders:** Custom Bootloader Development, Memory Mapping, Linker Scripts
- **Drivers & Interfaces:** SPI, I2C, CAN, LIN, Ethernet, UART, ADC, PWM, DIO
- **Debugging:** Lauterbach TRACE32 (PRACTICE), HIL, In-Vehicle Debugging
- **Tools:** Git, Jira, CMake, Jenkins, Docker, Vector CANoe/CANalyzer
- **Model-Based Design:** MATLAB, Simulink, Stateflow
- **Testing:** Unit Testing, Integration Testing, HIL, CI/CD Integration

EDUCATION

Bachelor in Electronics & Telecommunication Engineering

University of Pune, India

LANGUAGES

- **English** - C1 (Fluent)
- **German** - Telc B1 (Can easily make conversations)